# Post-Meeting Dry Lab Plan

Workstream Specifications Following Mentor Review

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The following seven workstreams were identified during the 03/07/26 mentor meeting as the priority computational tasks for the dry lab. Each section states the objective, scientific rationale, proposed approach, required data inputs, and expected deliverable.

## Task 1: Bayesian Protease Landscape Model

**Objective.** Replace deterministic threshold-based disease calls with a probabilistic model that outputs $P(\text{disease} \mid \text{signal})$ for each protease in the panel.

**Rationale.** A single protease concentration measurement carries uncertainty from biological variability, assay noise, and the non-exclusive nature of protease elevation (MMP14 is elevated in cancer, RA, and fibrosis; specificity is a population-level claim, not a single-sample certainty). A Bayesian model encodes these distributions explicitly and outputs a calibrated probability rather than a binary threshold call, which is both scientifically defensible and more informative for a point-of-care device.

**Approach.**

1. Model protease concentration in healthy and disease populations as log-normal distributions: [protease] $\sim$ LogNormal$(\mu, \sigma^2)$, with $(\mu_H, \sigma_H)$ and $(\mu_D, \sigma_D)$ estimated from literature for each protease.
2. Set a prior on disease prevalence $P(D)$ appropriate to the screening context (e.g. $P(\text{CRC}) \approx 0.004$ for a general population screen).
3. Apply Bayes' theorem:
$$P(D \mid c) = \frac{P(c \mid D)\, P(D)}{P(c \mid D)\, P(D) + P(c \mid H)\, P(H)}$$
where $P(c \mid D)$ and $P(c \mid H)$ are the log-normal likelihoods evaluated at measured concentration $c$.
4. Propagate sensor noise (electrochemical signal uncertainty) through the concentration estimate to produce a full posterior distribution on the disease probability.
5. For a multi-protease panel, combine per-protease posteriors under a conditional independence assumption (first pass) or a joint model if correlation data are available.

**Data sources.**

- Healthy and disease concentration distributions: literature per protease (Task 4 table feeds directly here).
- Disease prevalence priors: published epidemiology per indication.
- Sensor noise characterisation: electrochemical geometry model (Task 3).

**Deliverable.** Per-protease $P(D \mid c)$ curves with 95 % credible intervals; panel-level composite posterior; Python implementation reusable for any new protease added to the panel.

## Task 2: Hill Equation Dose-Response (Replace Linear Graph)

**Objective.** Refit the existing signal-vs-concentration plot with a Hill (sigmoidal) function to correctly represent sensor behaviour across its full operating range.

**Rationale.** The current linear graph misrepresents two physically real regimes: at sub-LOD concentrations the signal does not decrease linearly from baseline (a noise floor exists below which no change is detectable), and at high concentrations the response saturates as all surface-bound sensor molecules are cleaved. A linear fit drawn through the dynamic range implicitly extrapolates an invalid response at both extremes, which is immediately apparent to a reviewer familiar with biosensor characterisation.

**Approach.** For a signal-off electrochemical sensor, the Hill model is:

$$S(c) = S_{\max}\left(1 - \frac{c^n}{K_d^n + c^n}\right)$$

where $S_{\max}$ is the uncleaved baseline current, $K_d$ is the half-maximal cleavage concentration, $n$ is the Hill coefficient ($n = 1$ for a standard Michaelis-Menten regime), and $c$ is protease concentration.

1. Fit $K_d$ and $n$ to available experimental data; prior to wet-lab results, plot the model curve using literature $k_{\text{cat}}/K_M$ estimates with uncertainty bands.
2. Annotate the plot with three labelled regions: (i) sub-LOD noise floor, (ii) linear dynamic range (the usable detection window), (iii) saturation plateau.
3. Define LOD as the concentration producing a signal change of $3\sigma_{\text{noise}}$ above baseline, read off the fitted curve.

**Data sources.** MMP14 $k_{\text{cat}}/K_M \approx 5.5 \times 10^4$ $\text{M}^{-1}\text{s}^{-1}$ for SGRSENIRTA (literature); sensor noise from Task 3.

**Deliverable.** Updated dose-response plot replacing the current linear graph; one curve per protease once wet-lab $k_{\text{cat}}/K_M$ values are available.

## Task 3: Electrochemical Geometry Model (Marcus Theory)

**Objective.** Model the methylene blue (MB) to electrode electron transfer rate as a function of intact linker length, and use this to predict the sensor's baseline current and limit of detection.

**Rationale.** The electrochemical readout depends on efficient MB-to-electrode electron tunnelling in the intact (uncleaved) state. If the tether is too long, tunnelling rate drops exponentially and baseline current is unmeasurably small; too short and the construct may not fold correctly or the cleavage site may be sterically obstructed. The optimal linker length is not obvious without quantitative modelling.

**Approach.** The Marcus-theory tunnelling rate is:

$$k_{\mathrm{ET}} = k_0 \exp(-\beta d)$$

where $\beta \approx 1.1$ Å$^{-1}$ for protein-tethered redox labels on flexible peptide linkers, $d$ is the MB-to-electrode distance in the intact state (Å), and $k_0$ is the pre-exponential factor at contact.

1. Extract end-to-end distances of the intact construct as a function of linker residue count from the AlphaFold3 structure (existing CIF artifact).
2. Compute $k_{\mathrm{ET}}$ vs. linker length; identify the range keeping $k_{\mathrm{ET}} \gtrsim 10$ s$^{-1}$ (practical SWV measurement floor at standard frequencies).
3. Estimate LOD from surface MB density, $k_{\mathrm{ET}}$, electrode area, and noise floor: a higher $k_{\mathrm{ET}}$ gives a larger baseline current and thus a lower detectable concentration for a $3\sigma$ drop.
4. Model the ratiometric correction: a co-immobilised inert reference probe (same geometry, non-cleavable) provides a denominator that cancels monolayer drift and non-specific desorption artefacts.

**Data sources.** AlphaFold3 CIF (on record); published $\beta$ values for flexible peptide tethers; MB square-wave voltammetry parameters from literature.

**Deliverable.** $k_{\mathrm{ET}}$ vs. linker length curve; predicted LOD as a function of surface coverage density; recommended linker length range for electrode fabrication by wet lab.

## Task 4: Clinical Threshold Mapping

**Objective.** For each protease in the panel, establish the clinically validated concentration range separating disease-positive from healthy samples, and verify that the sensor's predicted LOD (Task 3) falls below the disease threshold.

**Approach.** Literature search per protease and sample matrix; populate the table below. The detection feasibility column is computed as: LOD $<$ disease threshold $\Rightarrow$ feasible. Concentrations are reported as the range found in the literature; uncertainty in these values feeds directly into the prior distributions for Task 1.

| **Protease** | **Indication** | **Matrix** | **Healthy** | **Disease / Threshold** |
|---|---|---|---|---|
| MMP14 | CRC / solid tumours | Serum | ∼9 pM | 32–70 pM (cancer range) |
| MMP-8 | Periodontitis | Saliva / GCF | <20 ng/mL | >60 ng/mL (PerioSafe threshold) |
| MMP-2/9 | Chronic wound | Wound fluid | Baseline | To be sourced |
| Neutrophil elastase | Chronic wound | Wound fluid | Baseline | To be sourced |
| Serine proteases (fecal) | IBD / IBD-IBS diff. | Fecal sup. | Low | To be sourced |
| FAP | CRC (stromal) | Serum / tissue | Low | To be sourced |

**Deliverable.** Completed table with units, references, and a feasibility verdict per row; this is the clinical validation argument for the wiki and for judge Q&A.

## Task 5: Sample Volume and Sensor Amount Calculation

**Objective.** For each protease and sample matrix, calculate the minimum clinical sample volume needed to contain a detectable amount of protease at disease-threshold concentration, and the minimum sensor protein required on the electrode.

**Approach.** Extend the existing MMP14 blood back-calculation to the full panel. For a sample volume $V$ at disease-threshold concentration $c_{\text{thresh}}$:

$$n_{\text{protease}} = c_{\text{thresh}} \times V$$

The sensor must cleave enough substrate to drop voltammetric current by $3\sigma_{\text{noise}}$. The rate equation for signal-off kinetics gives:

$$[\text{cleaved}](t) = [\text{sensor}]_0 \left(1 - \exp\left(-\frac{k_{\text{cat}}}{K_M} \cdot [\text{protease}] \cdot t\right)\right)$$

The minimum detectable $[\text{protease}]_{\text{reaction}}$ is back-calculated from a 0.1 nM cleaved threshold (established for MMP14) over a 30-minute assay window. Each protease requires its own $k_{\text{cat}}/K_M$, sourced from MEROPS prior to wet-lab measurement.

For reference, the MMP14 calculation established: minimum $[\text{MMP14}]_{\text{reaction}} \approx 10.1$ pM; at cancer-range serum levels (32–70 pM), a 13–32 $\mu$L finger-prick is sufficient.

**Deliverable.** Per-protease table: minimum sample volume, minimum sensor loading per electrode, and assay time window needed for a positive call at disease threshold.

## Task 6: Cross-Reactivity Specificity Matrix

**Objective.** Quantify the probability that non-target proteases co-present in the same biological matrix will cleave the intended substrate motif, grounding specificity claims computationally.

**Rationale.** The SGRSENIRTA motif was selected for MMP14 preference (79 % MMP14 vs. 4 % MMP2, 9 % MMP9), but the full sample matrix contains additional proteases not assessed in the original literature. Any off-target cleavage produces a false-positive signal, so specificity must be modelled across all co-present proteases before the Task 7 ML tool can generate trusted candidates.

**Approach.**

1. For each protease in the panel, enumerate co-present off-target proteases in the same matrix (e.g. serum contains thrombin, plasmin, cathepsins, kallikreins, MMP-2/9).
2. Query MEROPS cleavage frequency data for each substrate motif against each off-target. The MEROPS frequency score represents relative cleavage probability at each substrate position given a protease's known repertoire.
3. Construct a specificity matrix $M$ where $M_{ij}$ is the relative cleavage rate of motif $i$ by off-target protease $j$, normalised to the on-target rate.
4. Compute selectivity index: $SI_i = r_{\text{on-target}} / \max_j(r_{\text{off-target},j})$. The existing MMP14 benchmark of $SI \approx 8$ is the minimum acceptable threshold.

**Data sources.** MEROPS cleavage frequency tables; published $k_{\text{cat}}/K_M$ for major proteases per matrix.

**Deliverable.** Specificity matrix per sample matrix; selectivity index per motif; identification of any off-target risks requiring motif redesign. This matrix is the validation criterion for ML-predicted sequences in Task 7.

## Task 7: ML Substrate Peptide Prediction Tool

**Objective.** Build a software tool that takes a target protease sequence as input and returns a ranked list of 4–5 candidate ∼9-residue substrate peptides, optimised for cleavage selectivity over all co-present off-target proteases. Scope is fixed to the confirmed panel proteases for iGEM 2026.

**Rationale.** Current substrate selection (e.g. SGRSENIRTA for MMP14) relies on manual literature curation. A computational tool that generates and ranks novel motifs from sequence alone dramatically accelerates panel expansion and constitutes a standalone software deliverable for the Best Software Tool prize. The wet-lab validation loop ensures each predicted sequence is experimentally confirmed, closing the model-to-experiment feedback cycle that is a primary judging criterion.

**Pipeline.**

1. **Input.** Target protease amino acid sequence or UniProt accession. Optionally: list of co-present off-target proteases in the intended matrix.
2. **Training data.** All experimentally validated cleavage sites for the target protease extracted from MEROPS (P4–P4′ 8-residue windows around the scissile bond). Negative examples: sequences confirmed not cleaved by the target but cleaved by off-targets.
3. **Feature extraction.** ESM2 (650M parameter model) residue-level embeddings of each candidate 9-residue window. ESM2 captures evolutionary and physicochemical context without hand-engineered features and is available open-access from Meta.
4. **Classifier.** Lightweight MLP or attention head trained on ESM2 embeddings to predict cleavage probability. Two scores per candidate: (a) $P(\text{cleaved} \mid \text{target protease})$ and (b) $P(\text{cleaved} \mid \text{off-target}_j)$ for each matrix-relevant off-target.
5. **Ranking.** Selectivity index $SI = P_{\text{target}} / \max_j(P_{\text{off-target},j})$ is the primary ranking criterion. Confidence intervals are reported alongside each candidate.
6. **Output.** Top 4–5 ranked ∼9-aa motifs with $SI$ score, confidence interval, and a per-residue contribution summary. These sequences are passed directly to wet lab as cleavage assay substrates.
7. **Wet-lab feedback loop.** Measured $k_{\text{cat}}/K_M$ for each validated motif is appended to the training set as a labelled data point. SGRSENIRTA (MMP14, validated) is the seed training example.

**Data sources.** MEROPS cleavage data; ESM2 pretrained weights (Meta, open access); wet-lab cleavage assay results (iterative feedback).

**Deliverable.** Python CLI: `predict_substrate.py --protease <UniProt> --matrix <serum|saliva|wound|fecal>`; ranked output table; tool documented on wiki software page; source on team GitLab.

## Summary

| # | Task | Approach | Deliverable |
|---|---|---|---|
| 1 | Bayesian landscape | Log-normal priors, Bayes' theorem, panel posterior | $P(D \mid c)$ curves per protease |
| 2 | Fix dose-response | Hill equation replacing linear regression | Updated signal-vs-concentration plot |
| 3 | Electrochemical model | Marcus theory ($k_{\mathrm{ET}} \propto e^{-\beta d}$), AlphaFold3 distances | $k_{\mathrm{ET}}$ vs. linker length; predicted LOD |
| 4 | Clinical thresholds | Literature per protease / matrix | Threshold table with feasibility verdict |
| 5 | Sample volume | Rate equation back-calculation per protease | Min. sample volume and sensor loading |
| 6 | Cross-reactivity | MEROPS frequency data, selectivity index | Specificity matrix per matrix |
| 7 | ML peptide tool | ESM2 embeddings, MEROPS training data, selectivity ranking | `predict_substrate.py`; top 4–5 motifs |